Dynamic Time Warping

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```r
data_long <- read_csv("data_b_long.csv")
```

```
## Rows: 897390 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (6): industry_code, question, industry_name, question_name, question_ty...
## dbl  (2): level, value
## date (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
data_wide <- read_csv("data_b_wide.csv")
```

```
## Rows: 332 Columns: 2656
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl  (2654): C0000000_KLD, C0000000_GUS, C0000000_GES, C0000000_LUS, C000000...
## lgl     (1): C1020000_KLD
## date    (1): date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

1. DTW Clustering

1.1 Cluster Industries with Similar Temporal Evolution

If two industries exhibit similar dynamic patterns across all questions, they will be clustered together.

```r
dl_level1 <- data_long %>%  ## extract industries from level1
  filter(level == 1)

dw_level1 <- dl_level1 %>%  ## wide format
  select(date, industry_code, question, value) %>%
  tidyr::pivot_wider(names_from = question, values_from = value) %>%
  arrange(industry_code, date)

dw_level1_clean <- dw_level1 %>% ## drop NA
  group_by(industry_code) %>%
  drop_na() %>%
  ungroup()

ts_list <- dw_level1_clean %>%  ## build bivariate time series (wide format)
  group_by(industry_code) %>%
```

```r
  group_split() %>%
  lapply(function(df) {
    df %>%
      select(-industry_code, -date) %>%
      as.matrix()
  })
names(ts_list) <- unique(dw_level1_clean$industry_code)

## DTW clustering
set.seed(123)
cluster_result <- tsclust(
  ts_list,
  type = "partitional",
  k = 4,
  distance = "dtw_basic",
  centroid = "dba",
  trace = TRUE
)
```

```
## Iteration 1: Changes / Distsum = 22 / 1780357
## Iteration 2: Changes / Distsum =  2 / 1696652
## Iteration 3: Changes / Distsum =  2 / 1693351
## Iteration 4: Changes / Distsum =  0 / 1692620
##
##  Elapsed time is 6.9 seconds.
```

```r
cluster_result ## cluster size
```

```
## partitional clustering with 4 clusters
## Using dtw_basic distance
## Using dba centroids
##
## Time required for analysis:
##    user  system elapsed
##   39.48    0.23    6.90
##
## Cluster sizes with average intra-cluster distance:
##
##   size  av_dist
## 1    1     0.00
## 2    8 85787.90
## 3    5 85346.35
## 4    8 72448.14
```

```r
cluster_result@cluster ## cluster assignment
```

```
##  [1] 3 3 2 2 2 2 2 2 3 4 3 4 2 4 4 4 4 4 4 1 2 3
```

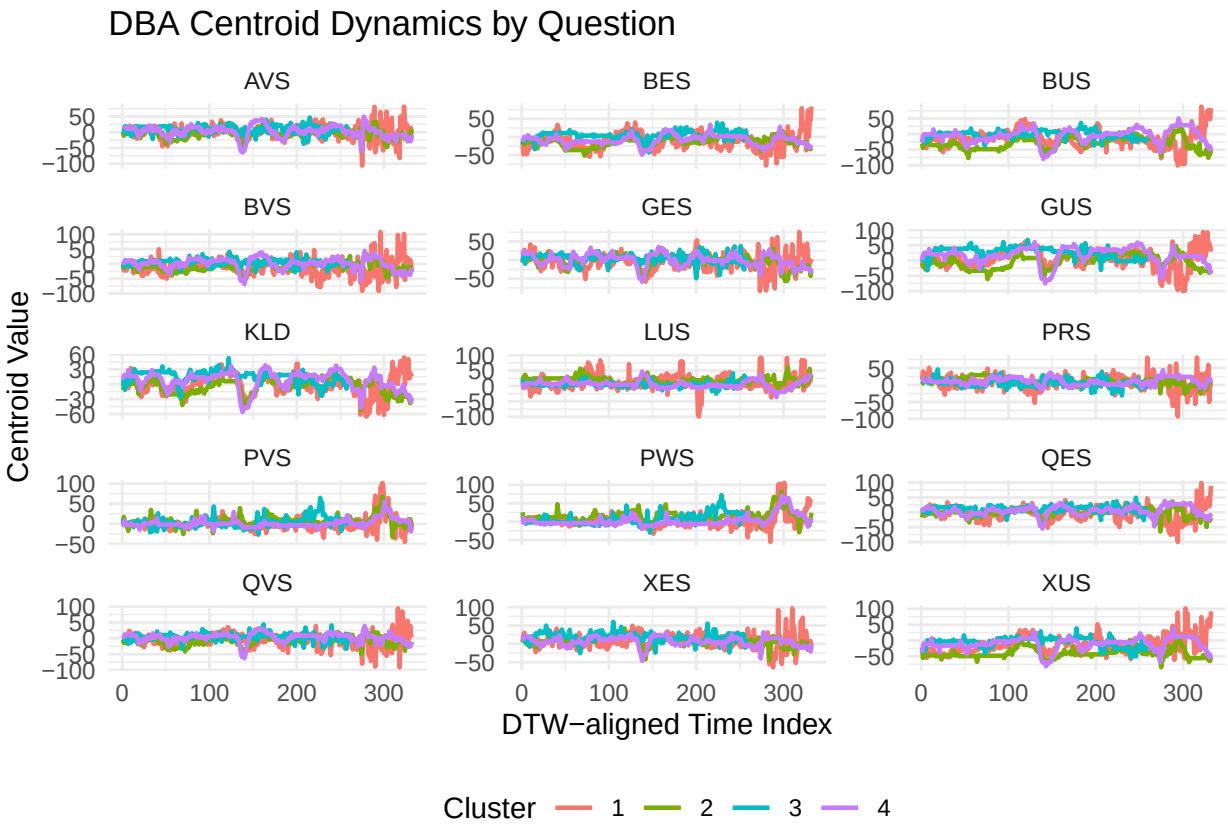
```r
industry_cluster <- tibble( ## cluster assignment table
  industry_code = names(ts_list),
  cluster = cluster_result@cluster
) %>%
  left_join(
    dl_level1 %>%
      distinct(industry_code, industry_name),
    by = "industry_code"
```

```r
  )
industry_cluster
```

```
## # A tibble: 22 x 3
##    industry_code cluster industry_name
##    <chr>           <int> <chr>
##  1 C1000000            3 Herstellung von Nahrungs- und Futtermitteln
##  2 C1100000            3 Getränkeherstellung
##  3 C1300000            2 Herstellung von Textilien
##  4 C1400000            2 Herstellung von Bekleidung
##  5 C1500000            2 Herstellung von Leder  Lederwaren und Schuhen
##  6 C1600000            2 Holz-  Flecht-  Korb- und Korkwarenherstellung (ohne M~
##  7 C1700000            2 Papiergewerbe
##  8 C1800000            2 Herstellung von Druckerzeugnissen
##  9 C1900000            3 Kokerei und Mineralölverarbeitung
## 10 C2000000            4 Herstellung von chemischen Erzeugnissen
## # ... with 12 more rows
```

```r
industry_centroids <- cluster_result@centroids ## cluster centroid
industry_centroid_long <- map2_df(
  industry_centroids,
  seq_along(industry_centroids),
  ~ {
    as_tibble(.x) %>%
      mutate(time = row_number(),
             cluster = .y) %>%
      pivot_longer(
        -c(time, cluster),
        names_to = "question",
        values_to = "value"
      )
  }
)

ggplot(industry_centroid_long,  ## plot DBA centroid
       aes(x = time, y = value, color = factor(cluster))) +
  geom_line(linewidth = 0.8) +
  facet_wrap(~ question, ncol = 3, scales = "free_y") +
  labs(
    title = "DBA Centroid Dynamics by Question",
    x = "DTW-aligned Time Index",
    y = "Centroid Value",
    color = "Cluster"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Results:

1. **Cluster assignment:** 4 clusters in total, industries within the same cluster exhibit similar dynamic patterns in survey-based business-cycle conditions (not similarity in production technology or sector definitions), allowing for time shifts across cycles.

2. **Cluter size:**

   - Small size (1): markedly different dynamic behavior, idiosyncratic or sector-specific dynamics. Potentially reflects: idiosyncratic industries (e.g. energy, automotive, highly export-dependent sectors); extremely strong reactions to the business cycle, may be lead/lag.

   - Medium size (5–8): shared and similar/core business-cycle dynamics while allowing for lead-lag structure, the type of non-obvious structure to be identified.

3. **DBA centroids: common business-cycle trajectory of each cluster.**

   We also compare the evolution of the same survey question across clusters. This allows us to isolate differences in timing, amplitude, and persistence of cyclical responses, and to assess which clusters and indicators may contain leading information for aggregate business conditions.

   → Axis:

   - **x-axis: the DTW-aligned time index**
     It reflects a common latent time scale obtained after dynamic time warping, allowing for lead–lag adjustments across industries. As a result, identical x-axis positions do not necessarily correspond to the same calendar date, but to comparable phases of the business cycle.

   - **y-axis: DBA centroid value of the respective indicator**
     Each centroid represents the average dynamic trajectory of all industries within a cluster, computed

under DTW alignment. The values correspond to standardized survey balances, with positive values indicating favorable assessments and negative values indicating pessimistic sentiment.

→ Characteristics:

- **Timing (lead–lag)**:
  Cluster 1 exhibits systematically earlier downturns in expectations-related indicators, suggesting potential leading-indicator properties for aggregate business sentiment. Cluster 4 does not exhibit systematic leading or lagging behavior, but rather phase-shifted or regime-specific responses.
- **Amplitude:**
  Cluster 1 includes the most sentiment-sensitive industries, showing amplified reactions to macroeconomic shocks. Cluster 3 corresponds to structurally stable industries whose sentiment indicators remain close to neutral throughout the cycle. Cluster 4 displays selective sensitivity: strong reactions in specific indicators, but muted responses elsewhere.
- **Persistence:**
  Cluster 1 captures highly cyclical industries with strong but short-lived responses to macroeconomic shocks. Cluster 2 displays high persistence of negative sentiment, indicating sectors with slower adjustment and prolonged weakness after downturns. Cluster 3 corresponds to structurally stable industries whose sentiment indicators remain close to neutral throughout the cycle. Cluster 4 is characterized by low persistence but pronounced episodic deviations.

→ **In summary:** Cluster 1 reacts early and strongly to shocks, making it a potential leading indicator. Cluster 2 exhibits persistent negative sentiment and slow recovery, while Cluster 3 represents more stable industries with muted cyclical responses. Cluster 4 represents industries with atypical, regime-dependent dynamics: neither clear leaders nor followers, but sectors that temporarily deviate from the main cycle, potentially signaling structural or shock-specific adjustments.

1.2 Cluster Questions with Similar Temporal Evolution

If two survey questions exhibit similar dynamic patterns across industries and over time, they may capture the same underlying economic mechanism, even if their wording or interpretation differs.

```r
dw_question <- dl_level1 %>%  ## wide format
  select(date, industry_code, question, value) %>%
  pivot_wider(
    names_from = industry_code,
    values_from = value
  ) %>%
  arrange(question, date)

dw_question_clean <- dw_question %>%  ## drop NA
  group_by(question) %>%
  drop_na() %>%
  ungroup()

ts_question_list <- dw_question_clean %>% ## build bivariate time series
  group_by(question) %>%
  group_split() %>%
  lapply(function(df) {
    df %>%
      select(-question, -date) %>%
      as.matrix()
  })
names(ts_question_list) <- unique(dw_question_clean$question)
```

```r
## DTW clustering
set.seed(123)
cluster_result_question <- tsclust(
  ts_question_list,
  type = "partitional",
  k = 4,
  distance = "dtw_basic",
  centroid = "dba",
  trace = TRUE
)
```

```
## Iteration 1: Changes / Distsum = 15 / 1464660
## Iteration 2: Changes / Distsum =  1 / 1509008
## Iteration 3: Changes / Distsum =  0 / 1503320
##
##  Elapsed time is 3.09 seconds.
```

```r
cluster_result_question
```

```
## partitional clustering with 4 clusters
## Using dtw_basic distance
## Using dba centroids
##
## Time required for analysis:
##    user  system elapsed
##   10.35    0.05    3.09
##
## Cluster sizes with average intra-cluster distance:
##
##   size   av_dist
## 1    3  99567.19
## 2    3  83060.11
## 3    5 105790.22
## 4    4 106621.63
```

```r
cluster_result_question@cluster ## cluster assignment
```

```
##  [1] 1 3 3 1 2 1 2 4 4 4 4 3 3 2 3
```

```r
question_cluster <- tibble(  ## cluster assignment table
  question = names(ts_question_list),
  cluster = cluster_result_question@cluster
) %>%
  left_join(
    data_long %>%
      distinct(question, question_name, question_type),
    by = "question"
  )
question_cluster
```

```
## # A tibble: 15 x 4
##    question cluster question_name                  question_type
##    <chr>      <int> <chr>                          <chr>
##  1 AVS            1 Auftragseingang                Comparison
##  2 BES            3 Beschäftigungserwartung        Expectation
##  3 BUS            3 Bestellungsbestand             Current
```

```
##  4 BVS            1 Nachfrage gegen Vormonat        Comparison
##  5 GES            2 Geschäftserwartungen            Expectation
##  6 GUS            1 Geschäftslage                   Current
##  7 KLD            2 Geschäftsklima                  Composite
##  8 LUS            4 Kapazitätsauslastung            Current
##  9 PRS            4 Produktionserwartung            Expectation
## 10 PVS            4 Verkaufspreise                  Comparison
## 11 PWS            4 Verkaufspreise Erwartung        Price
## 12 QES            3 Inlandsnachfrage Erwartung      Expectation
## 13 QVS            3 Inlandsnachfrage gegen Vormonat Comparison
## 14 XES            2 Exporterwartungen               Expectation
## 15 XUS            3 Auslandsauftragsbestand         Current
```

In contrast to industry clustering, the purpose of question clustering is not to identify leading or lagging behavior, but to uncover groups of survey indicators that exhibit similar cyclical dynamics across industries.

Results: The DTW clustering of survey questions yields four clusters. All clusters exhibit relatively large average DTW distances, indicating that even within clusters, questions are not identical but share similar temporal evolution up to phase shifts and scale differences.

→ **Economic interpretation of question clusters:**

- **Cluster 1:** Auftragseingang (AVS), Nachfrage gegen Vormonat (BVS), Geschäftslage (GUS).
  → Current-state dimension:
  Cluster 1 may indicate current business conditions and short-term demand changes. These questions are closely tied to firms' contemporaneous assessments of order inflows and overall business situation.
- **Cluster 2:** Geschäftserwartungen (GES), Geschäftsklima (KLD), Exporterwartungen (XES).
  → General expectations factor:
  Cluster 2 consists of forward-looking and composite indicators, combining expectations about future business conditions with export-oriented sentiment.
- **Cluster 3:** Beschäftigungserwartung (BES), Bestellungsbestand (BUS), Inlandsnachfrage Erwartung (QES), Inlandsnachfrage gegen Vormonat (QVS), Auslandsauftragsbestand (XUS).
  → Capacity-planning and workload dimension:
  Despite mixing current and expectation-based questions, Cluster 3 is the largest cluster and these indicators share a common dynamic pattern related to demand pressure and production planning.
- **Cluster 4:** Kapazitätsauslastung (LUS), Produktionserwartung (PRS), Verkaufspreise (PVS), Verkaufspreise Erwartung (PWS)
  → Cost–capacity–price channel:
  Cluster 4 is related to a supply-side adjustment mechanism, where firms respond to demand conditions through utilization, output planning, and pricing.

In summary: Rather than reproducing predefined survey categories, the clustering uncovers latent dynamic structures based on the temporal evolution of indicators. This provides a data-driven way to reduce dimensionality and select representative questions for monitoring business-cycle dynamics in practice, while avoiding over-interpretation of lead–lag relations at the question level.

Next step planned: **2. Network Based on DTW**

The DTW distance measures the dynamic similarity of the overall survey-question vector between two industries over the commonly observed months.